

Ice Navigation

Part 1 and Part 2

Ice Navigation Part 1: Ice navigation in theory, Unit 1, Exercise 1

1. A
2. C
3. C
4. A
5. B
6. A
7. C
8. B
9. A

Ice Navigation Part 1: Ice navigation in theory, Unit 2, Exercise 1

Ice report

VHF radio

Piping

Condition of tanks

Anchor windlass

Pilot ladder

Searchlight

Anchors

Propeller and rudder

Cooling water

Ice navigation Part 1: Ice navigation in theory, Unit 2, Exercise 2

find out	the channel used by the icebreaker
test	the searchlights
move astern	the anchors
drain	the pipes on deck
listen to	the daily ice reports
cover	anchor windlass
keep in a sheltered place	the pilot ladder
order	an ice chart
check	VHF radio

Ice Navigation Part 1: Ice navigation in theory, Unit 2, Exercise 3

pipes on deck	are	drained of water
mooring hawser reels	are	covered
cooling water	is	available
pilot ladder	is	not ice-covered
ballast tanks above the water-line	are	emptied of water
propeller	is	completely below the water surface

Ice Navigation Part 1: Ice navigation in theory, Unit 2, Exercise 4

VHF radio **has been checked**.

The pipes on deck **have been drained** of all water.

Containers **have been emptied** of any liquids.

All water **has been pumped** out from ballast tanks above the water-line.

The anchor windlass **has been protected**.

The pilot ladder **has been kept** in a sheltered place.

The searchlights **have been tested**.

The anchors **have been moved** astern.

Ice Navigation Part 1: Ice navigation in theory, Unit 3, Exercise 1

What do you do when you...

meet ice for the first time in your life?

<respect it>

meet thick and hard large floes?

<go around them, if possible>

meet ice which is obstructing?

<keep the ship moving>

meet a lead?

<follow it>

meet ridged ice?

<keep away because there is a lot more ice under the water>

get stuck?

<keep the propeller moving>

cannot move forward?

<set the rudder amidships and reverse>

Ice Navigation Part 1: Ice navigation in theory, Unit 4, Exercise 1

When a vessel **proceeds** in a convoy, a careful watch shall be **kept** for signals coming from the icebreaker or another vessel in the convoy. During hours of darkness, icebreakers **use** a fixed blue all-around light at the top of the mast. If the icebreaker **stops** or slows speed unexpectedly, two rotating red warning lights, **installed** one upon the other, are **lit**. The master of the vessel assisted shall **take** all possible measures to stop his vessel as quickly as possible.

A vessel in convoy shall **inform** the icebreaker without delay if she stops or slows her speed substantially. If the vessel stops due to the ice condition, the searchlight must be **switched off** for as long as the vessel **remains** stationary.

Ice Navigation Part 1: Ice navigation in theory, Unit 5, Exercise 1

1. True
2. False
3. False
4. False
5. False
6. False
7. True
8. True
9. False
10. True
11. True
12. False
13. False
14. True
15. False

Ice Navigation Part 1: Ice navigation in theory, Unit 6, Exercise 1

	icebreaker	
		towing cable
block		
		strop
		messenger wire
nock		
		bollard top
bollard		
		capstan
heaving line		
	anchor winch	
	vessel to be towed	

Ice Navigation Part 1: Ice navigation in theory, Unit 7, Exercise 1

1. False
2. True
3. False
4. True
5. False
6. True
7. False
8. False
9. False
10. True

Ice Navigation Part 1: Ice navigation in theory, Unit 7, Exercise 2

common	usual
rough	stormy
relative	comparative
significant	important
appropriate	suitable
available	accessible
hard	difficult

Ice Navigation Part 1: Ice navigation in theory, Unit 7, Exercise 3

Icing is quite a common **phenomenon** near the ice edge during low temperatures and rough sea. Icing takes place when the air temperature is below 0 degrees Celsius but the sea is not covered by ice. Icing is caused by the lifting of spray into the relative wind by the ship bow. The spray is then super-cooled and carried over the ship superstructure to freeze on bulkheads, decks, and rigging. The water splashing on the deck freezes to the superstructures of the ship. The weight of **accumulated ice** can be significant. It can **raise** the centre of the gravity of the ship and **deteriorate** the ship's stability. **Consequently/consequently**, the ship may even capsize, if ice is not **removed**.

In case of ice build up on the deck structures appropriate tools should be available. **Detaching/detaching** the ice cover is a hard job. Hammers, even sledge hammers and hardwood clubs can be used but wisely. When hitting with a sledge hammer on a thick layer of ice the load on the metal sheet **spreads** rather evenly into the surrounding surface. In case the ice layer is thin the blow **concentrates** onto a very small area and may force a dent into the plating. In such cases it is better to use heavy wooden clubs or even better leave it as it is.

Ice Navigation Part 1: Ice navigation in theory, Unit 8, Exercise 1

The **ice season** brings with it characteristic accident types. Ice can cause different kinds of **damage**. They vary from main engine malfunction due to heavy ice conditions to paint finish degradation due to **abrasion** by ice pieces. In most cases the damage is small and **limited** to shell plating damage between a few frames at most.

There are a few typical situations which can cause damage to vessels navigating through ice or in **an ice channel**. The vessel may not be able to **give way** due to ice and may **collide with** another vessel. Also, vessels navigating in a compressive ice field might get **stuck/beset** and sustain ice

damage to the **hull**. Compressive ice induces high local loads that exceed the strength requirements of plating and framing. A vessel stuck in the ice may also **drift** with the moving ice field and may eventually run aground.

When ships are moving in an ice channel **in succession** with a short lead, the first ship may get stuck in the ice and the next one may collide with it. **Backing/backing/Reversing/reversing** in ice may damage aft ship areas, especially the rudder and the propeller. In order to **avoid** getting stuck in ice, it is important that the propeller **maintains** its rotation. Usually the propeller ice loads are quite **short-term** and the so-called ice torque does not stop the propeller.

Ice Navigation Part 2: Ice navigation in practice, Dialogue 1: In a convoy, Exercise 1

Calling an icebreaker

- Ice-breaker Kontio, Kontio, Kontio, this is Marina, Marina, Marina, call sign **MWYA3**.
- Marina, this is Kontio, good evening.
- Kontio, Marina. My position is **latitude 59** degrees and **49** minutes North, **longitude 024** degrees and **18** minutes East and we are **bound for** Kotka. Is the **ice situation** very bad there? Are you going to **assist** us?
- Marina, Kontio. We are **close to** Helsinki lighthouse. We are going **westward** with three other **westbound** ships towards south of Porkkala. After that we are coming to assist you. Stand by on **channel 08**. Proceed to the position **10 miles south of** Helsinki lighthouse and wait for further instructions.
- Kontio, Marina. **Well understood/well understood**. You are close to Helsinki lighthouse, going west with 3 ships **towards** south of Porkkala. After that you are coming to assist us. I will **stand by** on channel 08.

Ice Navigation Part 2: Ice navigation in practice, Dialogue 1: In a convoy, Exercise 2

We are going	westward
Stand by	on channel 08
We are going to pass you	on your port side
Keep	full ahead
We will take care of	the distance
We are	five miles west of you
You will be number 2	in the convoy
Follow	Marina

Ice Navigation Part 2: Ice navigation in practice, Dialogue 2: Towing, Exercise 1

1. Apu, this is Marina.
2. Marina, Apu.
3. Apu, Marina. I am ready for towing. My crew is on the forecastle head. We are ready to receive the towing lines.
4. Marina, Apu. Okay. Stop your speed.
5. Apu, Marina. Okay. We will stop our speed immediately.
6. Apu, Marina.
7. Marina, Apu.

8. Apu, Marina. I have stopped now. I am not moving.
9. Marina, Apu. Okay. You are not moving.
10. Marina, Apu. Dead slow ahead.
11. Apu, Marina. Dead slow ahead.
12. Marina, Apu. Keep slow ahead and keep our masts in the same line. Let go towing lines now.
13. Apu, Marina. Slow ahead, masts in the line. We are letting go towing lines.
14. Apu, Marina. We have let go towing lines now, the bow is clear.
15. Marina, Apu. Keep full ahead and follow us about 9 miles more.
16. Apu, Marina. Full ahead. We will follow you about 9 miles more.

Ice Navigation Part 2: Ice navigation in practice, Dialogue 2: Towing, Exercise 2

- Icebreaker Apu, Apu, Apu. This Marina, Marina, Marina.
- Marina, Apu.
- Apu, Marina. We are **ice-bound** in position latitude **59** degrees and **56** minutes North, longitude **025** degrees and **36** minutes East.
- Marina, Apu. Understood. Please **prepare** for towing. We are going to **tow** you in about 20 minutes.
- Apu, Marina. Okay. We will prepare for towing. You are going to tow us in about 20 minutes.
- Marina, Apu. **Keep full ahead/keep full ahead** while preparing; we will tell you when to **slow down**.
- Apu, Marina. Well received. **We will keep full ahead/we will keep full ahead**.

Apu reverses in front of Marina's bow.

- Marina, Apu.
- Apu, Marina.
- Marina, Apu. **Keep slow ahead/keep slow ahead**.
- Apu, Marina. **Slow ahead**.
- Marina, Apu. We are going to give you **two lines**.
- Apu, Marina. You are going to give us **two lines**. Do we need a **heaving line**?
- Marina, Apu. Yes, you need a **heaving line**.

- Apu, Marina. Towing lines are **made fast**.
- Marina, Apu. Towing lines fast. Ask your crew out of the **forecastle**. We are starting to move. Make half ahead. **Steer/steer** with us and keep our masts in the **same line**.
- Apu, Marina. Crew is clear of the **forecastle**. We are starting to move. We will make half ahead and steer with you. We keep our masts in the same line as yours.

- Marina, this is Apu.
- Apu, Marina.

- Marina, Apu. We will **finish** towing in about 20 minutes.
- Apu, Marina. Okay. Towing **will be finished** in 20 minutes.

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- Marina, Apu.
- Apu, Marina.

- Marina, Apu. We will finish assistance in about 10 minutes. You must proceed **northeastwards** alone. Are you ready to get the **waypoints**?
- Apu, Marina. Understood. **Assistance/assistance** will be finished in 10 minutes. We are ready to take the waypoints.